(6) Lubricated and install disc without threaded hole from Trac-Lok® Tool C-4487 into lower side gear.

(7) Install upper side gear and clutch disc pack with retaining clips seated in the case pockets (Fig. 59).

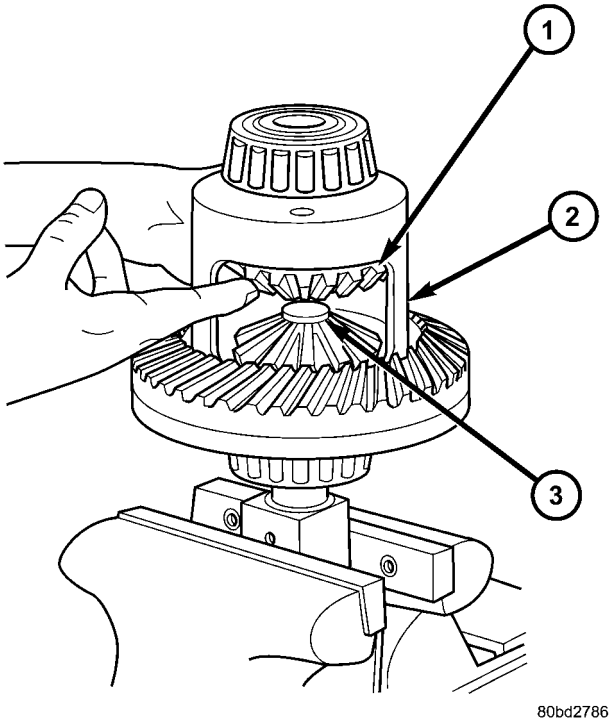

Fig. 59 CLUTCH PACK AND UPPER SIDE GEAR

1 - SIDE GEAR AND CLUTCH PACK
2 - DIFFERENTIAL CASE
3 - DISC

(8) Hold assembly in position and install threaded disc from Trac-Lok® Tool C-4487 into top side gear.

(9) Install forcing screw and tighten screw to slightly compress clutch disc.

(10) Place pinion gears in position in side gears and verify that the pinion mate shaft hole is aligned.

(11) Rotate case with turning bar C-4487-2 until pinion mate shaft holes in pinion gears align with holes in case. It may be necessary to slightly tighten the forcing screw in order to install the pinion gears.

(12) Tighten forcing screw to 122 N·m (90 ft. lbs.) maximum to compress the Belleville springs.

(13) Lubricate and install thrust washers behind pinion gears and align washers with a small screw driver. Insert mate shaft into each pinion gear to verify alignment.

(14) Remove Forcing Screw, Step Plate and Threaded Adapter.

(15) Install pinion gear mate shaft and align holes in shaft and case.

(16) Install pinion mate shaft roll pin. Stake (peen) case over pin in two places 180 degrees apart.

(17) Lubricate all differential components with hypoid gear lubricant.

DIFFERENTIAL - TRU - LOK

DESCRIPTION

The differential is a locking differential, that provides a positive mechanical connection between the right and left axle when engauged. The differential uses a dog clutch to connection the right and left axle.

OPERATION

The Tru-lok differential is activated by the axle lock switch located on the dash panel. When the switch is activated, an air pump with a built-in pressure regulator sends 5 PSI of air pressure to a accuator diaphragm in the differential housing. The diaphragm then engages a dog clutch and a position switch. The dog clutch has one gear attached to the differential case and another gear attached to a differential side gear. When the dog clutch is engaged the right and left wheels turn at the same speed. The position switch lights a lamp on the dash to indicate the system has been engaged. The differential works as standard differential when not engaged.

NOTE: The differential is serviced as an assembly, the diaphragm and indicator switch are serviced separately. The differential case must be removed to service the diaphragm actuator and indicator switch.

DIAGNOSIS AND TESTING

UNLOCKED

(1) Block tires opposite the axle to be tested to prevent the vehicle from rolling.

(2) Place transfer case in 4WD Low and automantic transmission in Park (1st gear if manual transmission).

(3) Raise both wheels of the axle to be tested off the ground.

(4) Turn ignition to the ON position and dash switch to the OFF position.

(5) Rotate one tire by hand. The other tire should spin in the opposite direction.

NOTE: If wheel cannot be rotated the differential must be repaired/replaced.

LOCKED

(1) Block tires opposite the axle to be tested to prevent the vehicle from rolling.